

SYSTEM DIAGRAM UPDATE NOTES

M.O.A.C. v3.1 — Architecture Changes from v3.0

June 13, 2026 · Melvin E. Bellinger Jr.

Note: This document is a working reference for updating the M.O.A.C. system architecture diagram. It is not a standalone technical disclosure. All substantive capability descriptions are contained in the accompanying Technical Disclosure and Implementation Summary documents.

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BASELINE: v3.0 DIAGRAM STRUCTURE

The v3.0 diagram is organized into four tiers:

| Tier | Label | Contents | |---|---|---| | 1 | Core Intelligence | LLM, Cloud routing, Vision, TTS | | 2 | Persistent Memory | Profile, Concept Store, Episodic, Relational, Rule Store | | 3 | General Reasoning Engines | Transfer, MCTS Planner, Zero-Shot Rules, Metacognitive Monitor, Recursive Self-Improvement | | 4 | Ambient Presence | Sensor Fusion, 5-Star Hierarchy, Proactive Speech, Guarded Agency |

The changes below describe additions and modifications required to bring the diagram current for v3.1.

TIER 2 ADDITIONS — PERSISTENT MEMORY

Add two new boxes to the Persistent Memory tier:

Episodic Store

- **Module:** `persistent-episodic-memory.js`
- **Sub-label:** "TF-IDF similarity retrieval · 2,000 episode capacity"
- **Connections:** Core Intelligence (read: pre-query; write: post-response)
- **Visual indicator:** Hot-path marker (see Legend section)

Semantic Fact Store

- **Module:** `persistent-episodic-memory.js` (same module as Episodic Store)
 - **Sub-label:** "Confidence-weighted · 500 fact capacity · cosine deduplication"
 - **Connections:** Episodic Store (same module), Core Intelligence (read: pre-query)
-

TIER 3 ADDITIONS — GENERAL REASONING ENGINES

Add three new boxes to the Reasoning Engines tier:

AGI Core Pipeline

- **Module:** `agi-core-pipeline.js`
- **Sub-label:** "Metacognition · Causal Reasoning · Theory of Mind · Explainability · Self-Preservation"
- **Position:** First in the tier — this module executes before all others on the hot path
- **Connections:** Core Intelligence (injects context pre-query)
- **Visual indicator:** Hot-path marker — this fires on every standard query turn
- **Color:** Visually distinguish from on-demand modules (suggested: accent border color)

Continual Knowledge Graph

- **Module:** `continual-knowledge-graph.js`
- **Sub-label:** "3,000 nodes · 8,000 edges · 6 typed relation types · cross-domain transfer"
- **Connections:**
 - Core Intelligence (read: pre-query; write: post-response)
 - Cross-Domain Transfer engine (feeds it)

- **Visual indicator:** Hot-path marker

Curiosity and Goal Engine

- **Module:** `curiosity-goal-engine.js`
 - **Sub-label:** "Novelty decay · Gap detection · Autonomous goal generation · Curiosity pulse"
 - **Connections:**
 - Autonomous Goal Engine (replaces and augments the prior box)
 - Core Intelligence (injects context pre-query)
 - **Visual indicator:** Hot-path marker
-

TIER 3 MODIFICATIONS — EXISTING BOXES

Autonomous Goal Engine

Update to reflect replacement by the new Curiosity and Goal Engine:

- **Old sub-label:** "goal-engine.js · self-generated goals · life-cycle"
- **New sub-label:** "curiosity-goal-engine.js · novelty tracking · gap detection · 20-goal capacity"

Metacognitive Self-Monitor

Update to reflect migration into the AGI Core Pipeline:

- **Old sub-label:** "confidence tracking · self-correction"
 - **New sub-label:** "agi-core-pipeline.js · per-query execution · confidence [0, 1] · knowledge boundary detection"
-

SELF-PRESERVATION LAYER UPDATE

The v3.0 diagram includes a "SELF-PRESERVATION LAYER" annotation at the bottom. Extend its description:

Add to existing description: " · per-query disk viability check · per-query heap memory monitor (`agi-core-pipeline.js`) · 30-second health cache · warns model of own resource constraints via prompt injection"

DIAGRAM LEGEND UPDATE

Add the following entries to the diagram legend:

| Indicator | Meaning | |---|---| | ⚡ Yellow border | **Hot-path module** — executes on every standard query turn | | Blue border | **On-demand module** — executes via API call or background task |

Existing legend colors (Foundation / Goal-tracking / Agency / Ambient) remain unchanged.

Modules requiring hot-path indicator (⚡):

- AGI Core Pipeline
- Episodic Store (read path)
- Continual Knowledge Graph (read path via `buildGraphContext()`)
- Curiosity and Goal Engine (read path via `buildCuriosityContext()`)

The remaining Tier 3 engines fire on-demand via API endpoints or scheduled background tasks and do not carry the hot-path indicator.

DATA FLOW ADDITIONS

Add the following to the diagram's data flow annotations:

Per-query (pre-LLM):

```
[Each query]
→ AGI Core Pipeline
  → metacognition result (confidence score + boundary directive)
  → causal frame (causal type + reasoning directive)
  → user mind model (emotion + expertise + communication directive)
  → health status (disk + memory warnings if applicable)
→ Episodic Store (read: relevant past episodes + semantic facts)
→ Knowledge Graph (read: related concepts by edge weight)
→ Curiosity Engine (read: curiosity pulse + active goals)
→ [All injected into volatile context → LLM]
```

Post-response (background, after HTTP response sent):

```
[After each response]
→ Episodic Store (write: new episode)
→ Knowledge Graph (write: new concepts + typed relations)
→ Curiosity Engine (write: novelty score update + gap detection + goal generation)
→ Semantic Fact Store (write: facts extracted from response)
```

Diagram reference document — not a standalone disclosure. — Melvin E. Bellinger Jr., June 13, 2026